Kuei-Hao (Howard) Li, Ph.D.

李 奎 皓



Ph.D., International Intercollegiate Ph.D. Program, National Tsing Hua University

Co-founder of **Uedu.tw**

Interaction Design for AI Systems · Human–AI Interaction · Conversational & Multimodal Interfaces · User-Centred Evaluation · English-Medium Instruction (EMI) Educator

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AREAS OF EXPERTISE

Human–AI Interaction & Conversational Interaction Design

Designing dialogue flows, turn-taking, and Socratic scaffolding-and-fading policies that let an AI assistant support users while preserving their agency and effort; published in the *International Journal of Human–Computer Interaction*.

User-Centred Design & Evaluation

User-perception studies, quasi-experimental comparisons, mixed-methods designs, and large-scale analysis of interaction logs (60,000+ turns) to evaluate how people actually use interactive systems.

Multimodal & Physiological Interaction

Integrating consumer wearables (Garmin partnership) and heart-rate-variability signals with conversational interfaces for affect-aware, adaptive system behaviour (PALM).

Immersive & Virtual Environments

Interaction-task design and user engagement in virtual-reality and virtual learning environments (CUE model; IEEE ICALT, EuroCALL).

Prototyping to Deployed Product

Full-stack implementation in JavaScript and Python, from interactive prototype to production service: the Uedu platform is deployed at ten universities and was shown in the EC-TEL 2026 Demonstration Track.

English-Medium Instruction (EMI) Instructor

Since 2016, has taught human–computer interaction, computer-mediated communication, and digital instructional design entirely in English at four national universities, with teaching excellence awards; able to deliver a full interaction-design curriculum in English.

RESEARCH SUMMARY

My research concerns the design, deployment, and empirical evaluation of agentic information systems built on large language models. It is organised around a single question of information-systems inquiry: *when an autonomous system assumes part of a user's cognitive work, how should that system be architected, instrumented, and governed so that the user retains agency over the task?* Addressing this question requires work at three connected levels, and my publications span all three: the architecture of multi-agent conversational systems; the data infrastructure required to instrument such systems at scale; and the evaluation of their behavioural effects through controlled studies and large-sample analytics.

This programme is conducted through Uedu Academy, an agentic AI learning platform that I co-founded and continue to develop in collaboration with National Central University. The platform serves a dual function: as a production service deployed across institutions in Taiwan and internationally, indexing more than 31,000 courses; and as a research instrument that generates de-identified multimodal interaction data under an IRB-approved gov-

ernance framework. This dual role allows pedagogical interventions to be studied at scale in authentic classroom settings rather than in laboratory conditions. I have established a research partnership with Garmin Taiwan, obtaining authorised API access for physiological data collection, and have worked with Microsoft to design and build AI systems for digital learning in higher education.

> Work drawing on this infrastructure has appeared in *Computers and Education: Artificial Intelligence* (JCR Q1; IF 23.4), the *International Journal of Human–Computer Interaction* (SCIE/SSCI Q1; IF 7), and *Frontiers in Psychology* (SSCI Q1; IF 4.2), and in the proceedings of ACM Learning @ Scale (Best Paper shortlist), EC-TEL, and several IEEE conferences. The platform received the **Technology Innovation Award, Bronze Prize**, at e-Learning Forum Asia 2026.

EDUCATION

2026 **Ph.D., International Intercollegiate Ph.D. Program**
National Tsing Hua University (NTHU), Hsinchu, Taiwan
Dissertation: *A Human-Centered Agentic AI Learning Framework for Higher Education: Design, Evaluation, and Learning Analytics*
Advisor: Prof. Tzu-Hua Wang · Dissertation defended 30 September 2026

RESEARCH AGENDA

I **System architecture for agentic AI.** Design principles for multi-agent LLM systems in which task planning, adaptive feedback, and reflective prompting are separated into cooperating components; orchestration policies that govern how much of a task the system performs; retrieval-augmented grounding in organisational knowledge bases; and on-premise inference as a response to institutional data-sovereignty constraints.

II **Instrumentation and data infrastructure.** Architecture and governance of multimodal data lakes integrating dialogue text, speech, behavioural logs, and wearable physiological signals; automated semantic classification of interaction traces at scale; de-identification, consent management, and the ethical reuse of behavioural data collected in operational systems.

III **Evaluation of user and organisational effects.** Quasi-experimental and large-sample observational designs for assessing system impact; mixed-effects modelling of nested interaction logs; adoption and acceptance of generative AI across disciplinary contexts; and AI literacy and governance as determinants of responsible use.

ACADEMIC AND PROFESSIONAL APPOINTMENTS

2022 – present **Co-Founder and Lead Developer, Uedu Academy**
National Central University (NCU), Taiwan

- Co-architect, with the founding principal investigator, of a production multi-agent system comprising learner-profiling, task-planning, adaptive-feedback, and self-reflection agents, deployed at ten universities.
- Develop and maintain the supporting infrastructure: retrieval-augmented knowledge bases, a public developer API and Model Context Protocol server, containerised computational services, and a self-hosted model served on campus GPUs for data-sovereign inference.
- Direct the research data pipeline underlying the platform's published studies, including a classified corpus of more than 60,000 interaction turns from 116 courses across four institutions.
- Lead the Uedu–Garmin Health research infrastructure, with authorised API access for physiological data collection in a learning analytics platform.

Ongoing **Co-Principal Investigator, AI and Digital Database Application Laboratory**
Industrial Technology Research Institute (ITRI), Taiwan

- Industry–academia programme on applied artificial intelligence and digital database systems; concurrently appointed ITRI instructor for professional training in AI application and development.

Project Manager
Microsoft, Taiwan

- Designed and built AI systems for digital learning in higher education, in cooperation with ITRI.

2018 – 2021 **Lecturer, Department of Education and Learning Technology**
National Tsing Hua University (NTHU)
English-medium instruction in cloud computing, blockchain applications, and digital instructional design. Outstanding Teaching Award.

2016 – 2018 **Lecturer, Department of Foreign Languages and Literatures**
National Yang Ming Chiao Tung University (NYCU; formerly National Chiao Tung University)
Computer-assisted language learning, computer-mediated communication, human–computer interaction, and academic and technical English.

2016 – 2018 **Lecturer, Language Center**
National Taiwan University of Science and Technology, National Taiwan University, and National Taiwan Normal University
English for science and technology; English for specific purposes; academic writing. Outstanding Teaching Award.

PUBLICATIONS

Author order follows each byline; the author's name is set in bold. Complete record at ORCID 0009-0007-3474-8489.

Refereed Journal Articles

J3 **Li, K.-H.**, & Chang, C.-K. (2026). Designing and evaluating a Socratic-scaffolded conversational AI assistant for educational tasks: Cross-cohort effectiveness, user perception, and the effort signature. *International Journal of Human–Computer Interaction*. https://doi.org/10.1080/10447318.2026.2717500
Taylor & Francis · SCIE & SSCI, Q1, IF 7 · First and corresponding author

J2 **Li, K.-H.**, & Chang, C.-K. (2026). Agentic AI as a dual-role lecturer and teaching assistant: Effects on learner autonomy, self-regulated learning, and intrinsic motivation in university-level EFL education. *Frontiers in Psychology, 17*, 1847607. https://doi.org/10.3389/fpsyg.2026.1847607
SSCI, Q1, IF 4.2 · First author

J1 Chang, C.-K., & **Li, K.-H.** (2026). Chat as learning: Student–AI conversations as discipline-associated cognitive engagement patterns. *Computers and Education: Artificial Intelligence*. https://doi.org/10.1016/j.caeai.2026.100644
Elsevier · JCR Q1, IF 23.4 · CiteScore 43.3

Refereed Conference Proceedings

C9 Chang, C.-K., & **Li, K.-H.** (2026). AI teaching assistants at scale: Cross-disciplinary patterns of adoption and cognitive engagement across hundreds of university courses. In *Proceedings of the ACM Conference on Learning @ Scale (L@S 2026)*, Seoul, Republic of Korea. https://doi.org/10.1145/3774398.3811596
Full paper · Shortlisted for the Best Paper Award

C8 Chang, C.-K., & **Li, K.-H.** (2026). PALM: Scaling physiologically-aware AI tutoring through consumer wearables and large language models. In *Proceedings of the ACM Conference on Learning @ Scale (L@S 2026)*, Seoul, Republic of Korea. https://doi.org/10.1145/3774398.3811576
Work-in-progress paper

C7 Chang, C.-K., **Li, K.-H.**, Cheng, C.-L., & Lin, T.-C. (2026). From wearables to classrooms: A person-centered feasibility study of HRV-based physiological monitoring for learning analytics. In *Proceedings of the IEEE International Conference on Swiss Data Science (SDS 2026)*. https://doi.org/10.1109/SDS70563.2026.00035

C6 Chang, C.-K., & **Li, K.-H.** (2026). Uedu: A six-dimensional educational omics platform bridging AI, physiological sensing, and classroom practice. *European Conference on Technology Enhanced Learning (EC-TEL 2026)*, Industry and Practitioner Track.

C5 Chang, C.-K., & **Li, K.-H.** (2026). Uedu: A mindfully designed AI-assisted learning platform across four universities in Taiwan. *European Conference on Technology Enhanced Learning (EC-TEL 2026)*, Demonstration Track.

C4 Chang, C.-K., & **Li, K.-H.** (2025). Designing an educational omics data lake: A multimodal infrastructure for technology-enhanced learning. In *Proceedings of the 7th International Conference on Modern Educational Technology (ICMET 2025)*, Kyushu University, Fukuoka, Japan. https://doi.org/10.1109/ICMET67594.2025.11452000

C3 Yen, C.-C., Chang, Y.-Y., Chai, Y.-Z., Hsieh, P.-T., **Li, K.-H.**, & Chang, C.-K. (2025). Correlation analysis of conversational focus with learning experience in Python courses. In *Proceedings of the 7th International Conference on Modern Educational Technology (ICMET 2025)*, Fukuoka, Japan. https://doi.org/10.1109/ICMET67594.2025.11451926

C2 Chang, J.-K., **Li, K.-H.**, Li, J.-T., Yen, C.-J., Furqon, E. N., Cheng, C.-L., et al. (2025). Construction of the Uedu education omics data lake and multimodal research on speech, text, and neurophysiological signals. In *Proceedings of the 2025 TANET and NCS Conference*, Taiwan.

C1 **Li, K.-H.**, & Huang, Y.-C. (2017). Interaction tasks of the CUE model in virtual language learning environments. In *Proceedings of the 17th IEEE International Conference on Advanced Learning Technologies (ICALT 2017)*, Paris, France.

Conference Presentations

P5 **Li, K.-H.** (2017). Student engagement and motivation in the CUE model for virtual reality learning environments. *EuroCALL 2017*, University of Southampton, United Kingdom.

P4 **Li, K.-H.** (2017). Learning analytics in learning management systems: The E3 platform at National Chiao Tung University. *19th International CALL Research Conference*, University of California, Berkeley, United States.

P3 **Li, K.-H.** (2017). Student engagement in the flipped classroom within virtual language learning environments. *19th International CALL Research Conference*, University of California, Berkeley, United States.

P2 **Li, K.-H.** (2016). The flipped classroom: Undergraduate perceptions of social media use in writing instruction. *33rd International Conference on English Teaching and Learning*, National Changhua University of Education, Taiwan.

P1 **Li, K.-H.**, & Chen, R.-H. (2015). Examining indirect complaint discourse in a computer-mediated communication context. *12th Wenshan International Conference*, National Chengchi University, Taipei, Taiwan.

RESEARCH INFRASTRUCTURE AND SYSTEMS

Architecture — **Uedu agentic AI architecture.** A multi-agent system in which learner profiling, task planning, adaptive feedback, and reflective prompting are implemented as cooperating components over both commercial and self-hosted language models. The orchestration layer implements Socratic questioning and a scaffolding-fading policy, and annotates each turn with a cognitive-level classification, providing a continuous process trace of system use.

Data — **Educational omics data lake.** A six-dimensional multimodal research infrastructure integrating cognitive, linguistic, physiological–neural, social, environmental, and ethical data streams, unified under a de-identified, IRB-governed schema and released to external researchers under controlled access.

Sensing — **PALM: physiologically-aware language model tutoring.** Integration of consumer wearables with LLM-based tutoring, supporting real-time multi-subject physiological capture and event-level alignment for affect-aware adaptation of system behaviour.

Inference — **On-premise inference service.** A campus-hosted, Traditional-Chinese-optimised model with transparent load telemetry, demonstrating a privacy-preserving alternative to commercial cloud inference for institutions subject to data-residency constraints.

RESEARCH PROJECTS AND COLLABORATION

Ongoing — **Applications of Artificial Intelligence in Education and Industry** — Co-Principal Investigator
ITRI, AI and Digital Database Application Laboratory

2026 – present — **Uedu–Garmin Health Research Infrastructure Initiative** — Lead
Joint initiative with National Central University establishing consumer wearables as shared research infrastructure for multimodal behavioural analytics.

Ongoing — **International collaboration, Taiwan – Greece**
Joint project with Prof. Iraklis, Department of Informatics and Telematics, Harokopio University of Athens, on AI-supported instruction and teacher agency.

2016 – 2017 **MOE Multilingual and Cultural Competence Programme** — Co-Principal Investigator Ministry of Education cross-campus faculty community on English-medium curriculum development; concurrent project on integrating academic integrity into ESP curricula (NYCU).

HONOURS AND AWARDS

2026 **Technology Innovation Award, Bronze Prize (Third Place)**, e-Learning Forum Asia (eLFA) 2026 — *Uedu: A Ubiquitous-Learning Platform Realizing Educational Omics Across Institutions, Disciplines, and the Learner's Physiology* (C.-K. Chang & K.-H. Li; No. eLFA2026A24S5374).

2026 **Best Paper Award shortlist**, ACM Conference on Learning @ Scale (L@S 2026).

Teaching Excellence Award, National Tsing Hua University — EMI course evaluation ranked in the top 30% of the Department of Education and Learning Sciences.

Teaching Excellence Award, Language Center, National Taiwan University of Science and Technology.

Full International Conference Travel Grant, National Yang Ming Chiao Tung University — competitive funding for an oral paper presentation abroad.

Invited Speaker, Microsoft — lecture on cloud computing and blockchain applications for higher education.

2016 – 2017 **Co-Principal Investigator appointment**, Ministry of Education Cross-Campus English Teaching Community, Multilingual and Cultural Competence Programme.

INVITED TALKS AND PROFESSIONAL SERVICE

Jul 2026 **Invited Tutorial co-presenter**, IEEE International Conference on Service-Oriented System Engineering (CISOSE 2026), Fukuoka, Japan — "Building and Verifying AI Systems with Agentic AI." Certificate of Appreciation conferred.

Ongoing **Invited instructor**, ITRI professional training programme on AI application and development.

2017 **Presenter**, MOE Multilingual and Cultural Competence Programme, Joint Outcomes Conference, Taipei.

2016 **Invited speaker**, Cross-Campus English Teacher Pedagogical Enhancement Programme, Language Center, National Ilan University.

2016 **Adjudicator**, 19th English Speech and Writing Competition, NTU Alumni Association of Yilan County, National Ilan University.

TEACHING

Courses taught at NTHU, NYCU, and NTUST (English-medium instruction)

Introduction to Cloud Computing · Blockchain Applications · Introduction to Internet and Networking · Human–Computer Interaction · Computer-Assisted Language Learning · Computer-Mediated Communication · Learning Analytics in Learning Management Systems · Social Network Analysis · Digital Instructional Design and Application · Internet Literacy and Ethics · Generative AI: An Interdisciplinary Introduction · AI Applications in Business · Seminar on Learning Technology

Academic and professional English for graduate programmes

English for Science and Technology · Technical and Scientific Writing · Academic English Writing · Academic Writing and Communication for IS Research

Proposed offerings for a Department of Interaction Design

Human–Computer Interaction and User Experience · Conversational and AI Interaction Design · Multimodal and Wearable Interaction · Generative and Agentic AI for Designers · Interactive Prototyping with JavaScript and Python · Behavioural Data Analytics for UX · Research Methods in Interaction Design · Information Ethics, AI Governance, and Literacy. *All courses can be delivered entirely in English.*

RESEARCH SUPERVISION

Capstone	Co-supervised undergraduate capstone teams at NTHU and National Central University on deep reinforcement learning for car-following behaviour, FPGA and Verilog system implementation, and natural language processing for sentiment–market index analysis.
2026	Mentored student research teams using the Uedu platform in MOE Teaching Practice Research projects and NSTC undergraduate research projects.

METHODS AND TECHNICAL COMPETENCIES

Quantitative	Crossed random-effects GLMMs and multilevel modelling of nested interaction data; latent profile analysis; quasi-experimental design with control-group comparison; sequential explanatory mixed-methods design; validated instrument administration and psychometric screening (FLCAS, AMS-C, CPSES, PAS).
Computational	LLM application and agent architecture; prompt specification for system behaviour; retrieval-augmented generation; deep learning in PyTorch; multimodal transformers; NLP pipelines for large-scale text classification; knowledge-graph construction; physiological and radar signal processing.
Engineering	Python and JavaScript; full-stack platform development; REST and public API design; Model Context Protocol server implementation; containerised services; GPU-served model deployment; Verilog at project level.
Languages	Mandarin Chinese (native); English (full professional proficiency; English-medium instructor since 2016, qualified to deliver a complete information-management curriculum in English).

References available upon request · 推薦人資料備索 · Updated October 2026